

Shop Wise AI Cart

An AI-Powered Smart Grocery Shopping Assistant for Easier, Safer, and Smarter Supermarket Trips

Presented By:

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Team Members & Contributions

Project Lead & AI Engineer

Overseeing overall architecture, camera scanning integration, and model training pipelines for product identification.

Full-Stack Developer

Building user profile management, smart budget calculation engines, and interactive supermarket store navigation features.

UX & Safety Specialist

Designing allergy protection systems, dietary preference filter logic, and expiry tracking reminder notifications.





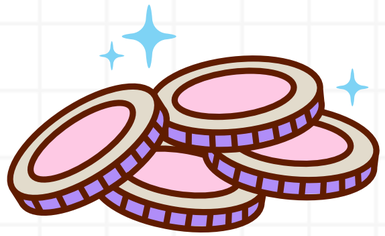
Idea Generation & Evaluation

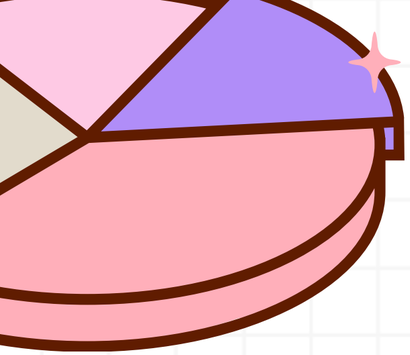
Initial Brainstorming Alternatives

We evaluated several standalone concepts during early ideation: 1) A barcode-only scanner app, 2) A static household budget tracker, and 3) A simple recipe recommendation tool based on fridge contents.

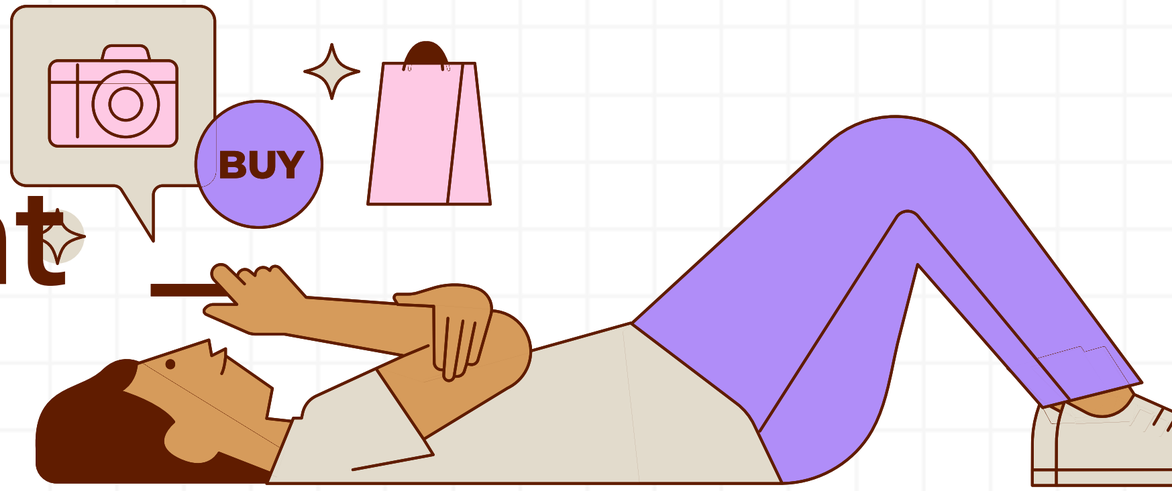
Why We Chose Shop Wise AI Cart

Users face fragmented experiences using multiple apps for budgets, diets, and allergies. Our evaluation proved that uniting all these capabilities into an AI-powered camera assistant offers unmatched convenience and market value.





Problem Statement & Target Users

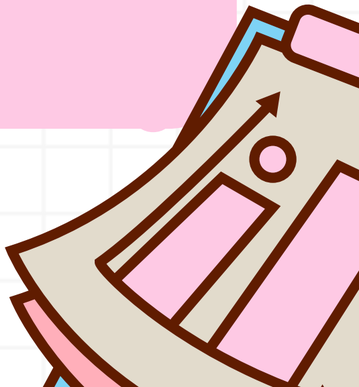


Everyday Supermarket Pain Points

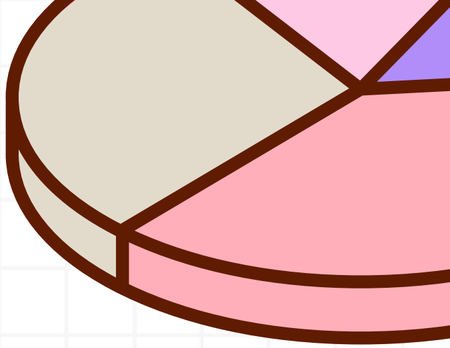
Shoppers frequently overspend their budgets, struggle to decipher complex ingredient lists with food allergies, waste hours navigating large hypermarkets like Carrefour or Lulu, and let purchased food expire unnoticed.

Target Audience & Impact

Targeted at busy families, budget-conscious individuals, and health-conscious shoppers with specific dietary restrictions (Halal, Keto, Vegan) or severe food allergies (peanuts, dairy).



Project Overview



All-in-One Solution: Combines camera recognition, budget control, allergy alerts, store navigation, and expiry reminders into a single unified app.

Supermarket Integration: Select specific stores like Lulu Hypermarket or Carrefour to match local inventory, aisle layouts, and pricing.

Personalized AI Guidance: Adapts to user profiles over time, learning shopping habits and making tailored product suggestions.



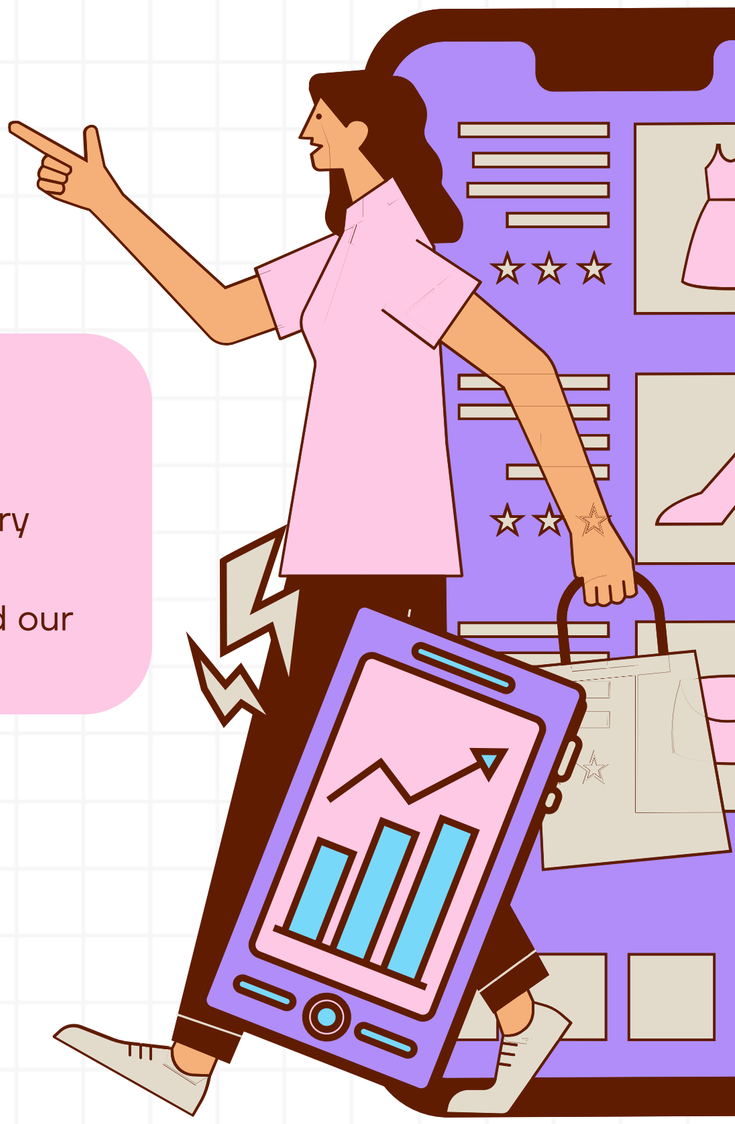
Conducted Research & Competitor Review

Market & Competitor Analysis

Existing supermarket apps offer basic online ordering or simple loyalty points, but lack real-time camera-based nutritional scanning, instant budget threshold warnings, and integrated allergy safety checks.

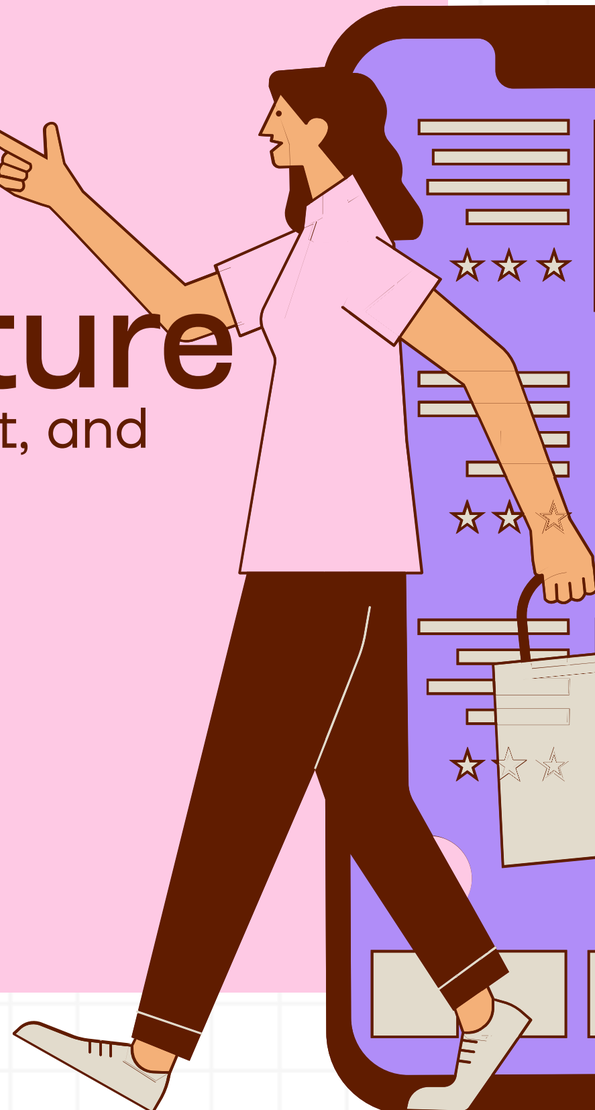
Key User Research Findings

Over 80% of surveyed shoppers reported accidentally exceeding their planned grocery budget and expressed anxiety over hidden allergens in packaged goods. This validated our core AI camera and smart budget features.



Part 2:Planning& Architecture

Data workflow, technical resource planning, risk assessment, and project scheduling.

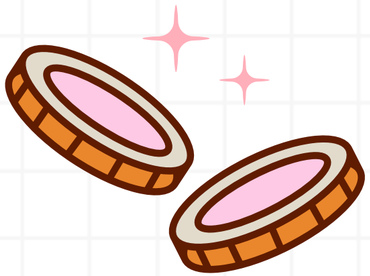


Data Analysis & User Journey

- **Profile Setup:** Users input budget limits, dietary preferences (Halal, Vegan, Keto), and allergy profiles.
- **Scan & Verify:** Camera AI matches visual input against product databases to evaluate prices, ingredients, and suitability.
- **Cart Optimization:** Real-time calculation monitors spending and suggests cheaper alternatives to keep within target budgets.



Resources & Technology Stack



AI & Vision Models

Computer vision libraries and image recognition models for real-time grocery product identification.

Database & APIs

Supermarket product inventory APIs (Lulu, Carrefour datasets), secure user profile storage, and recipe databases.

Development Tools

Mobile application frameworks, UI/UX design tools, and version control systems for seamless deployment.

Northline Group

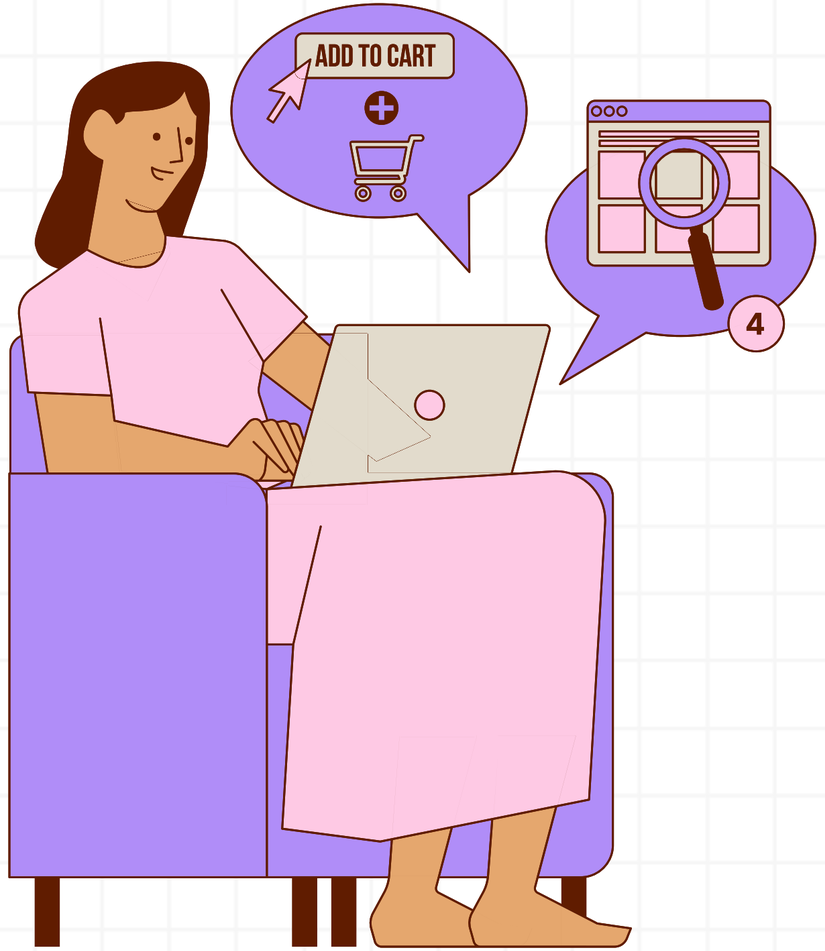
Challenges & Solutions

Technical Challenge

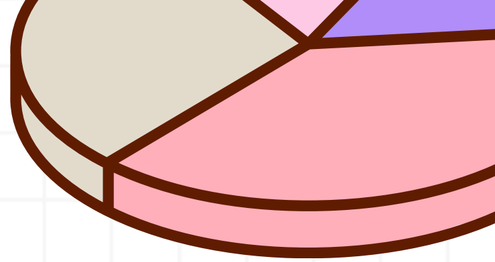
Accurately recognizing packaged grocery items under varying supermarket lighting conditions and blurry camera angles caused initial identification errors.

Implemented Solution

Accurately recognizing packaged grocery items under varying supermarket lighting conditions and blurry camera angles caused initial identification errors.



Risk Assessment & Mitigation



Identified Risk	Impact	Mitigation Strategy
Allergy Alert False Negatives	High (Safety)	Add mandatory disclaimer reminding users to verify actual product packaging.
Supermarket Price Changes	Medium	Implement periodic background API synchronization with store catalogs.
Usability & Complex UI	Medium	Keep camera scanning as the primary 1-tap interaction point.



Development Lifecycle & Timeline

Phase 1: Research

Requirements gathering, user surveys, and supermarket inventory mapping.

Phase 2: Design

UI wireframing, database schema creation, and AI model pipeline setup.

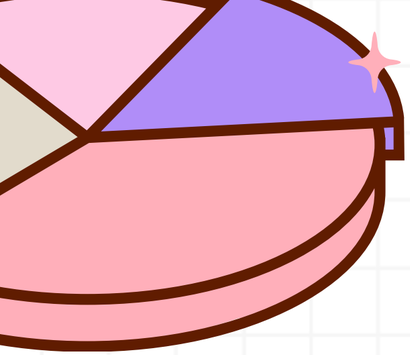
Phase 3: Launch

Core feature integration, prototype testing, bug fixing, and final deployment.

Part 3: Implementation & Future

Demonstrating AI execution, prototype results, future scaling plans, and conclusion.



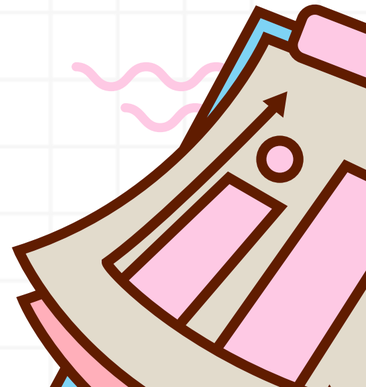


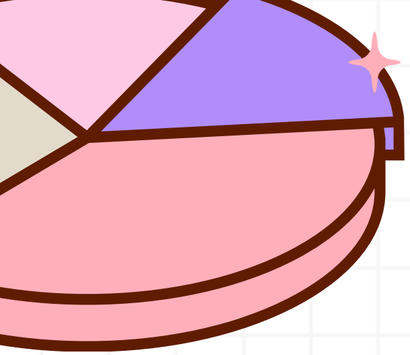
AI Implementation & Camera Scanner

Vision Workflow: Point phone camera at any item for instant image recognition and metadata retrieval.

Safety Cross-Check: Automatically compares ingredients with user allergy profile to trigger instant warnings.

Smart Recommendations: Suggests cheaper alternative products in real-time to save money.





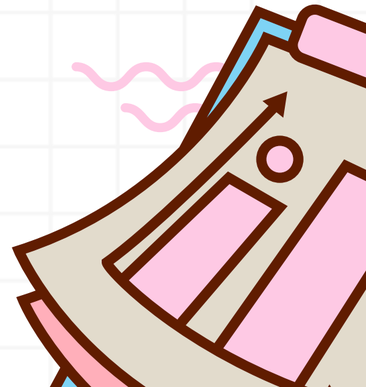
Initial Prototype vs. Final System

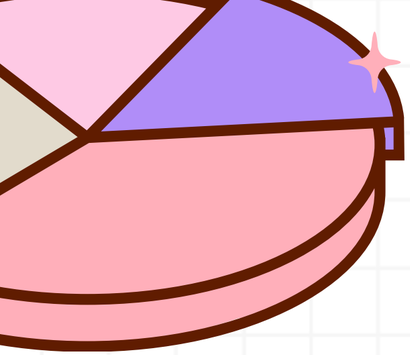
Initial Prototype

Focused purely on manual product search and basic budget summation, lacking automated image recognition, allergy warnings, and in-store aisle mapping.

Final Prototype

Delivers a complete, seamless assistant featuring camera AI scanning, live supermarket aisle navigation, expiry date tracking, and personalized recommendations.





Future Scaling & Enhancements

Optimized Routing

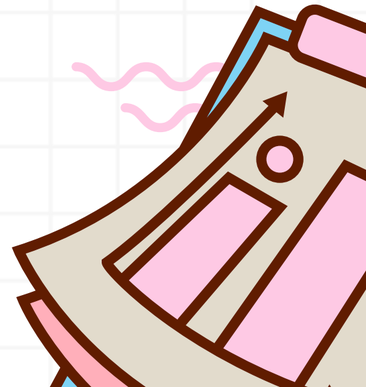
Generate efficient in-store walking paths based on full cart contents to eliminate backtracking.

Recipe Integration

Suggest recipes directly utilizing items nearing their expiry dates to minimize food waste.

Expanded Retailers

Integrate more supermarket chains and enable direct online checkout integration.



Thank You!

Shop Wise AI Cart — Making Grocery Shopping Smarter, Safer, and Easier.